Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **developer tooling**, **CI/CD automation**, and reliable **Kubernetes platforms** through concurrent processing, test infrastructure, monitoring, and measurable performance improvements.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under production load reliably daily.
- Achieved 99%+ production reliability across 5K+ monthly jobs through **data validation**, reconciliation checks, error handling, cloud storage integration, and CI/CD releases connecting FastAPI services and relational databases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive workflows during concurrent model execution at scale.
- Reduced feedback turnaround 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.
- Increased pipeline efficiency 68% by redesigning **AWS Glue and S3 ETL** workflows, automating JSON, CSV, and Parquet transformations for reliable API consumption and analytics across production environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Reduced resource wastage by 42% through **testing and production monitoring**, integrating automated suites, Docker/Jenkins delivery, Azure Monitor, Sentry, and Kibana across modular production services reliably at scale.
- Synchronized 2M+ records through an eight-service **real-time microservice platform** using React, TypeScript, Node.js, WebSockets, REST APIs, and Azure SQL across distributed operational workflows at scale consistently and reliably.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Streamlined project delivery by 22% through **Flask services and Firebase integration**, applying automated testing, real-time synchronization, and cross-platform workflows across mobile and web product environments consistently and reliably.
- Designed the system architecture and end-to-end workflow, mapped 12+ **backend integration points**, and presented the solution to stakeholders for alignment before implementation across transactional product services reliably.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, Bash, Shell, SQL
Platform & Cloud: Kubernetes, Docker, AWS, Azure, GCP, Linux, Microservices, Distributed Systems, REST APIs
CI/CD & Tooling: CI/CD, GitHub Actions, Jenkins, Git, GitHub, Automated Testing, Unit Testing, Integration Testing, PyTest
Reliability & Performance: Performance Optimization, Caching, Multithreading, Concurrency, Data Validation, Error Handling, Production Troubleshooting, Cloud Monitoring
Data & Engineering: PostgreSQL, MongoDB, Redis, AWS RDS, AWS S3, AWS Glue, ETL Pipelines, System Design

Projects

Monorepo Build Acceleration Platform - Design Concept

- Designed a **Go-based monorepo build platform** concept combining dependency graphs, content-addressed caching, incremental builds, parallel execution, reproducible containers, CI metrics, and reversible deployment gates for faster engineering feedback.

Agent-Aware CI Test Orchestrator - Design Concept

- Designed an **agent-aware CI test orchestrator** concept using Python, Kubernetes, GitHub Actions, test-impact analysis, flaky-test quarantine, concurrent sandboxes, structured context injection, and DORA metrics for measurable developer velocity.

FinNexus | Live

- Built a responsive **Kubernetes transaction platform** supporting 50K+ users and 1M+ transactions using TypeScript, MongoDB, WebSockets, secure APIs, real-time synchronization, advanced filtering, and CI/CD workflows at scale.